

Number of Prime Ministers Per Province/Territory Circuit

CP220 Project Phase III

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169037714

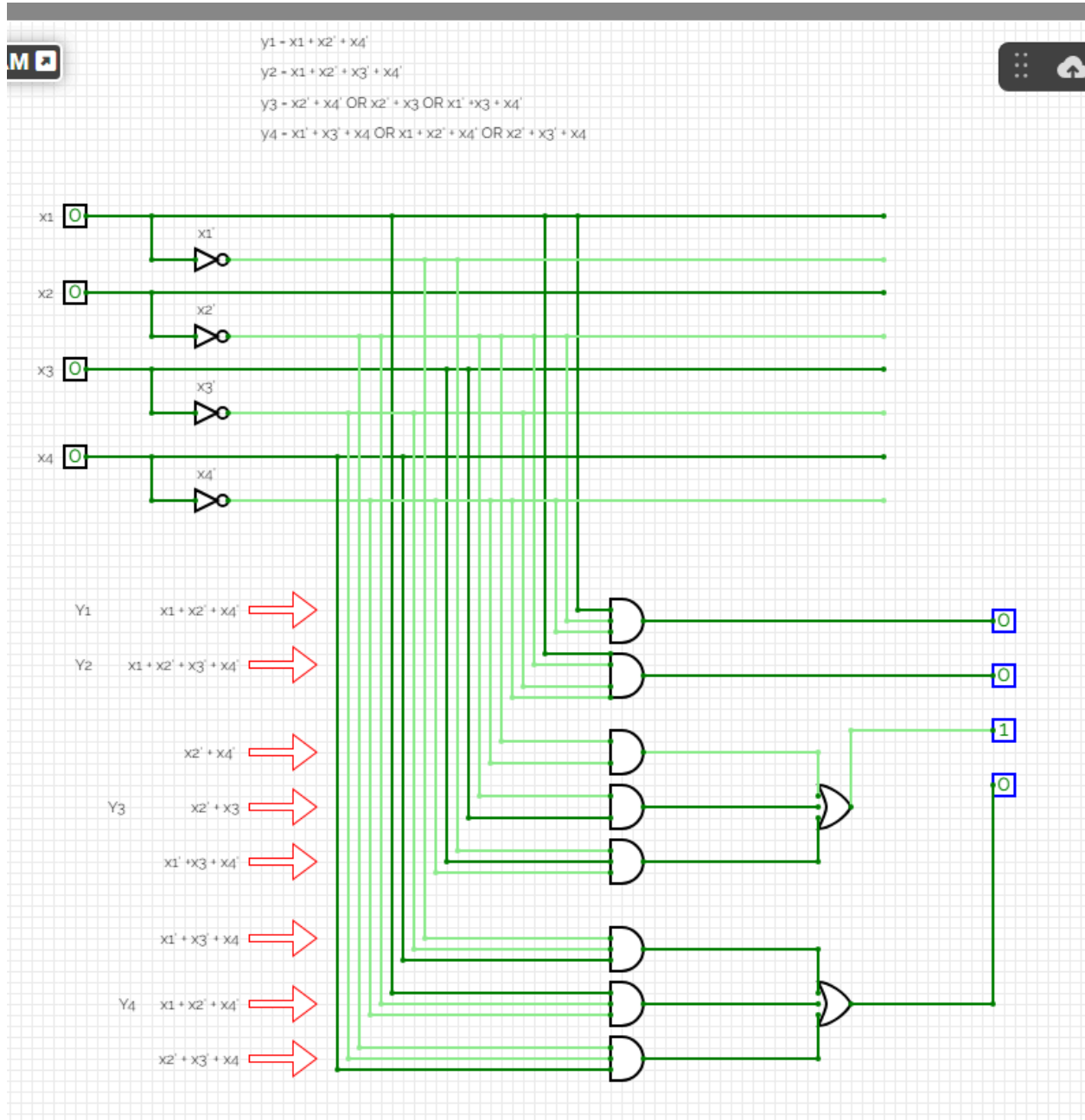
Fall 2023

Truth Table

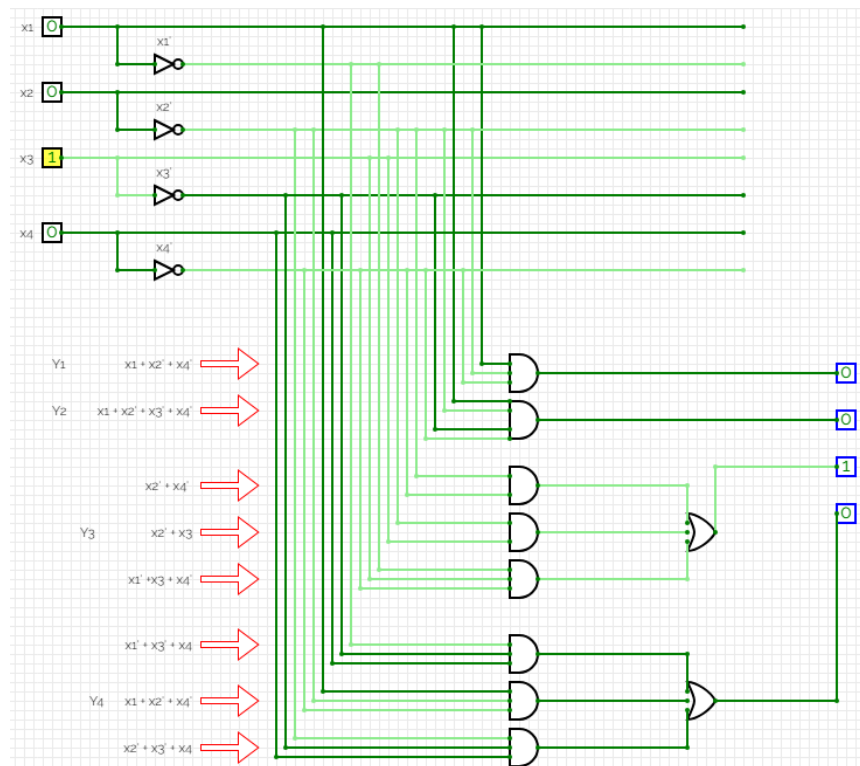
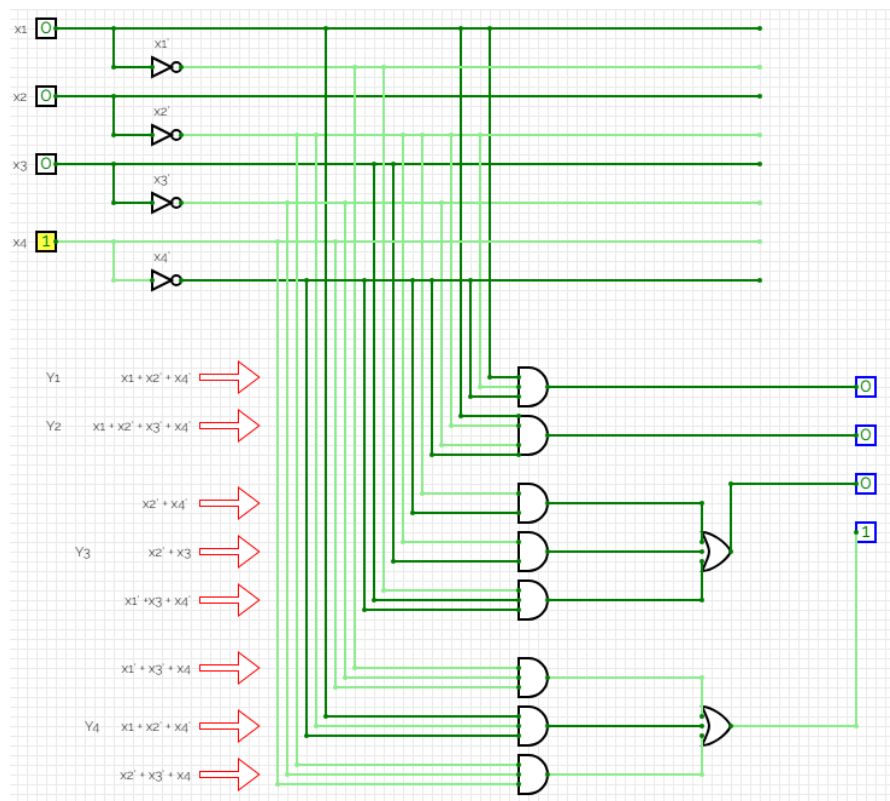
x_1	x_2	x_3	x_4		y_1	y_2	y_3	y_4
0	0	0	0		0	0	1	0
0	0	0	1		0	0	0	1
0	0	1	0		0	0	1	0
0	0	1	1		0	0	1	0
0	1	0	0		0	0	0	0
0	1	0	1		0	0	0	1
0	1	1	0		0	0	1	0
0	1	1	1		0	0	0	0
1	0	0	0		1	1	1	1
1	0	0	1		0	0	0	1
1	0	1	0		1	0	1	1
1	0	1	1		0	0	1	0
1	1	0	0		0	0	0	0
1	1	0	1		0	0	0	0
1	1	1	0		0	0	0	0
1	1	1	1		0	0	0	0

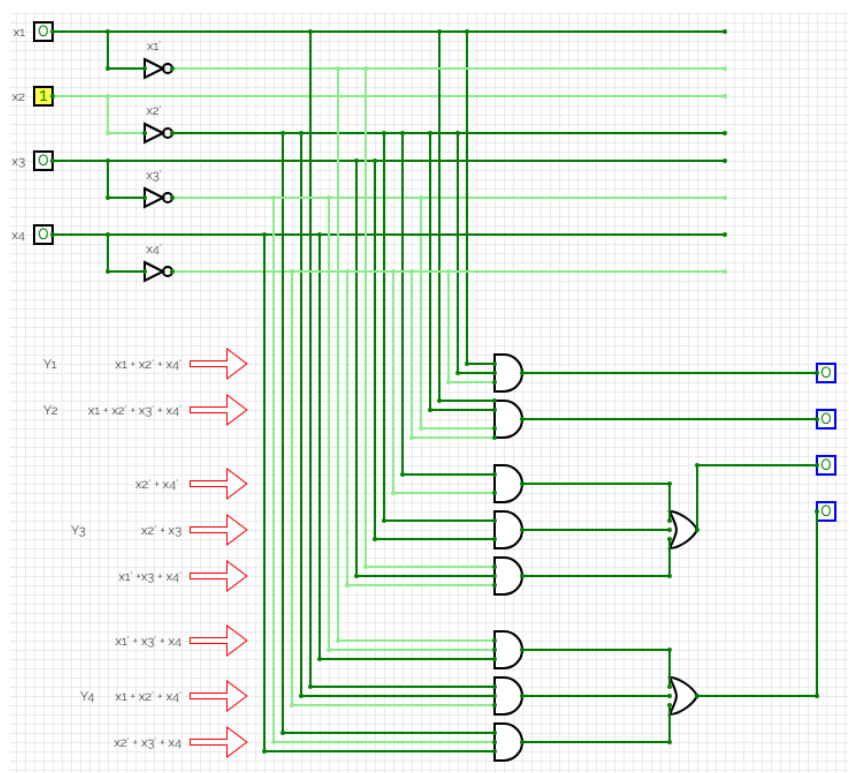
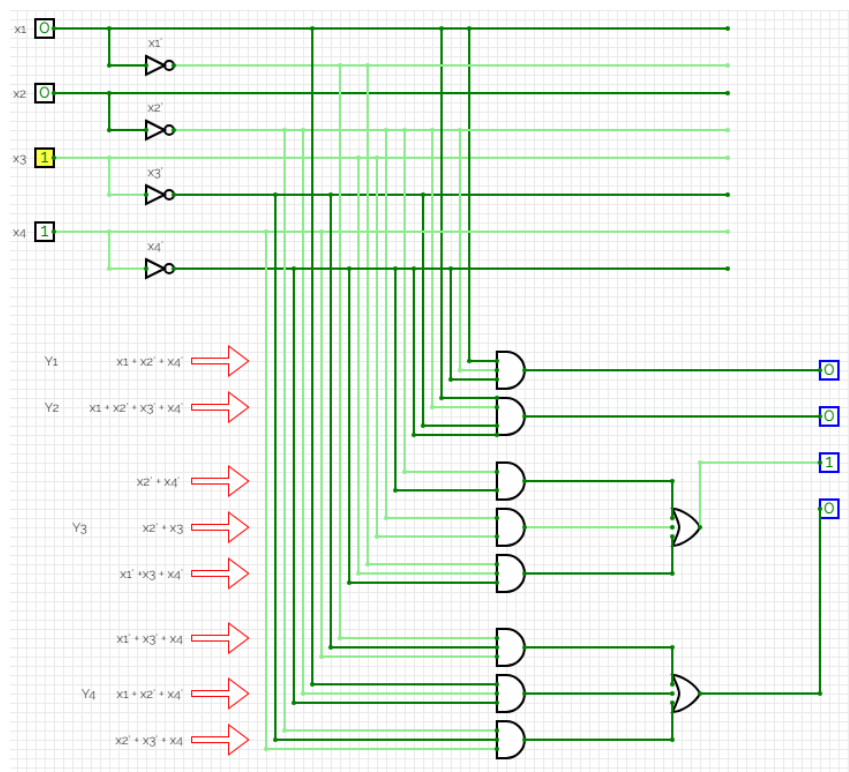
Outputs and Circuit

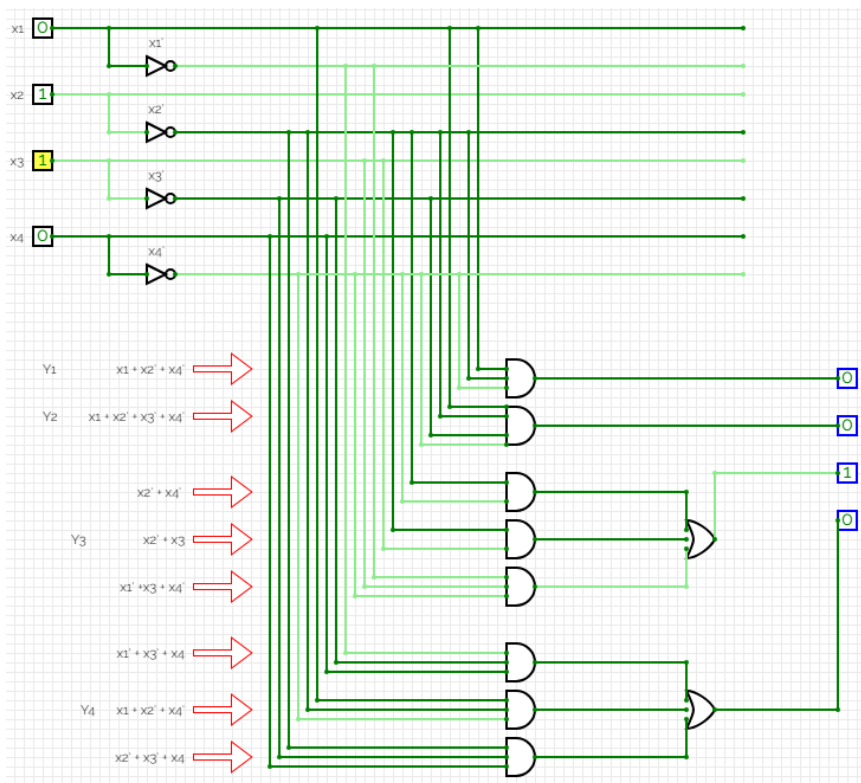
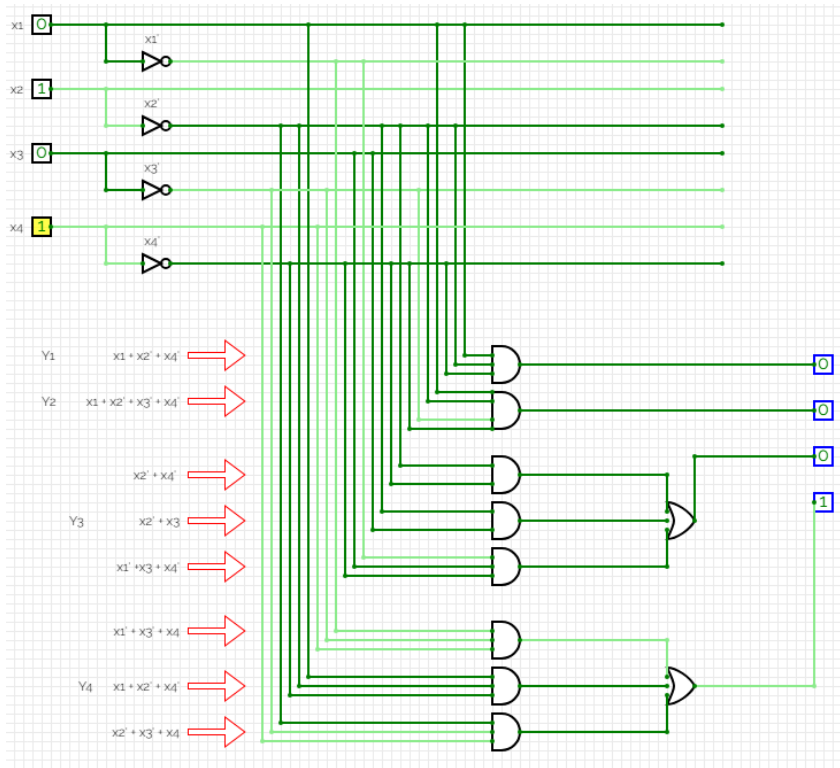
Here is the circuitverse for the Number of Prime Ministers Per Province/Territory Circuit:
As you can see the first output is already showing for Alberta with 2 Prime Ministers.

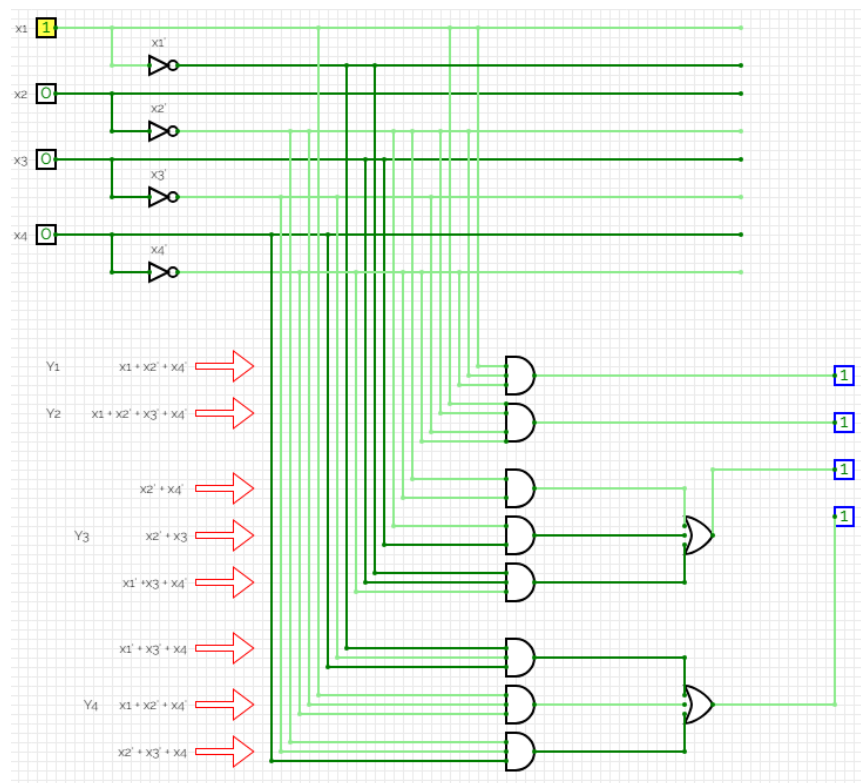
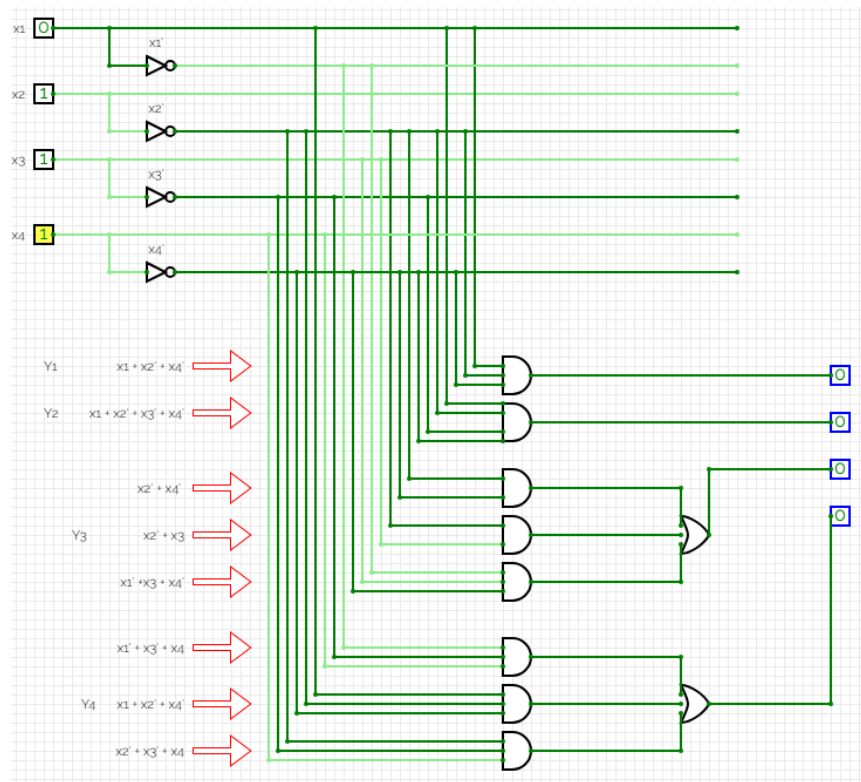


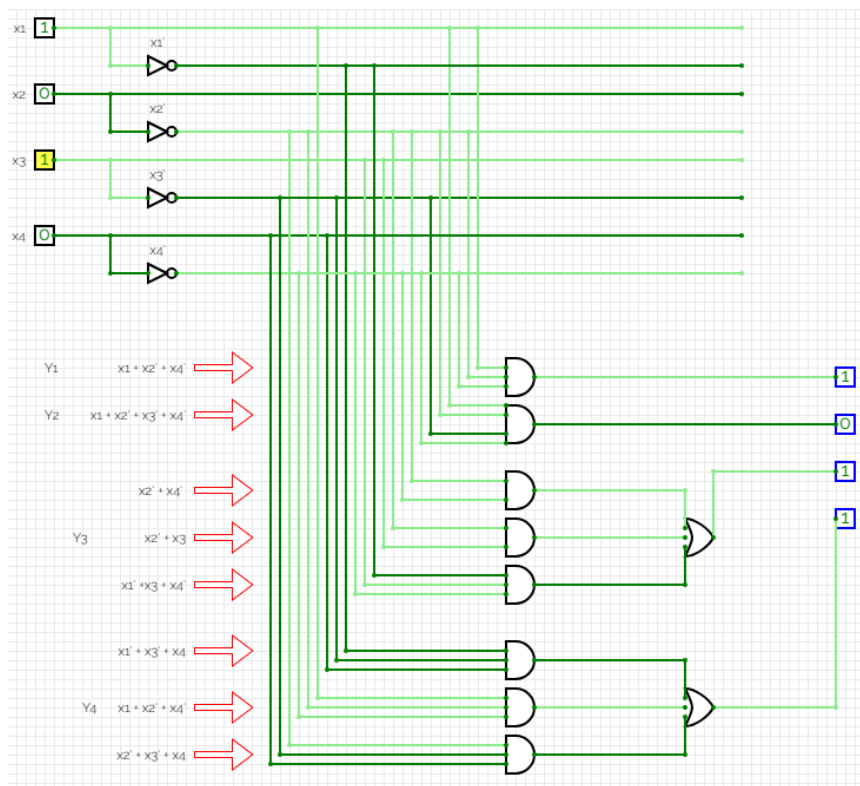
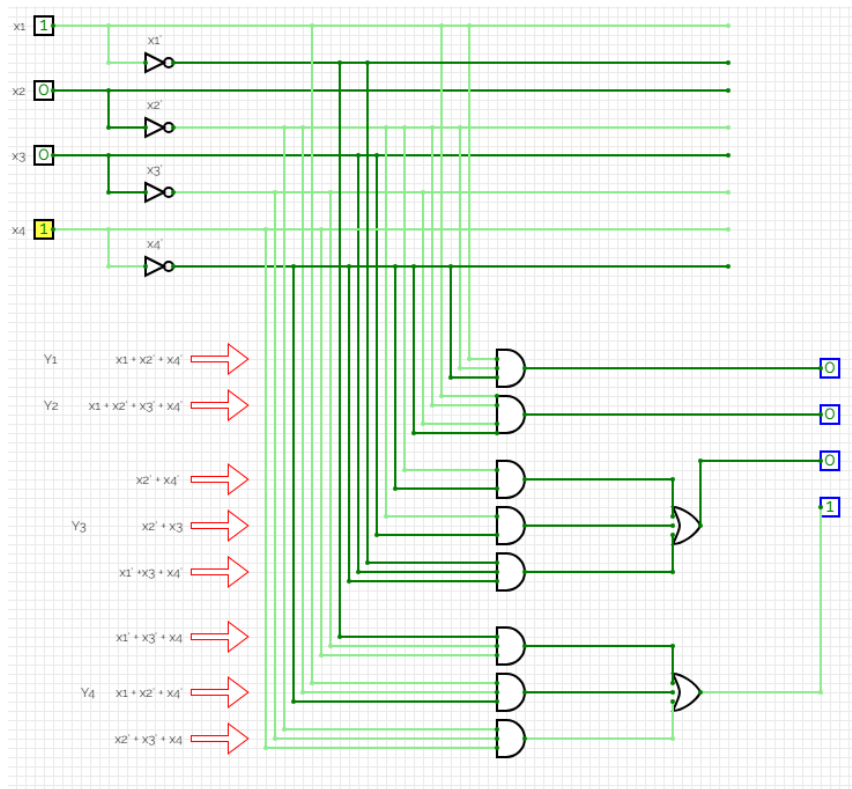
Now here are the rest of the outputs for each province as well as the final 3 which all output 0 as they are errors since there are no provinces or territories past 13:

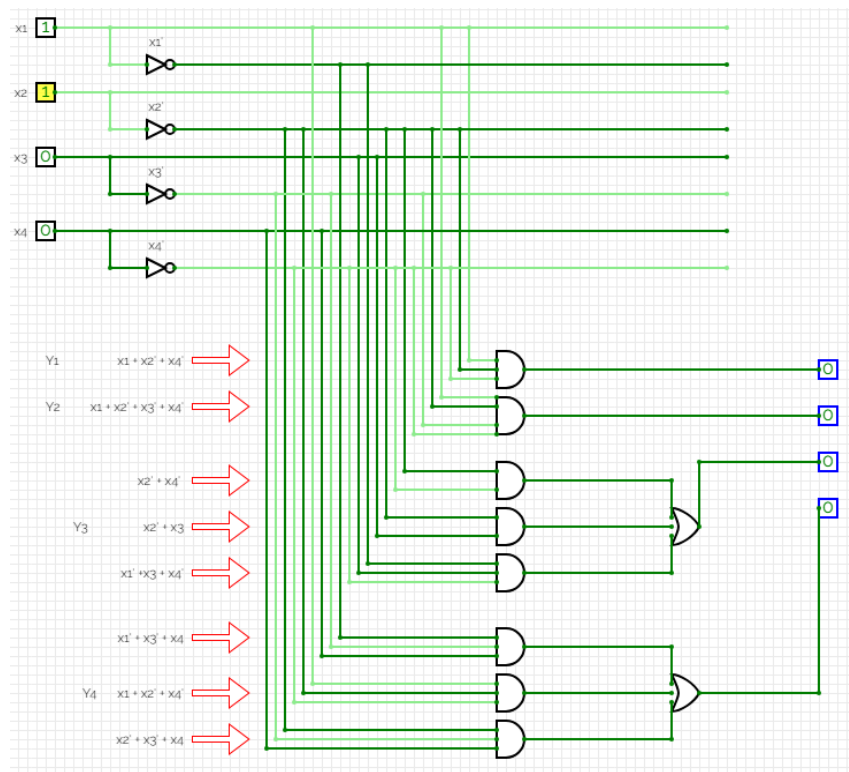
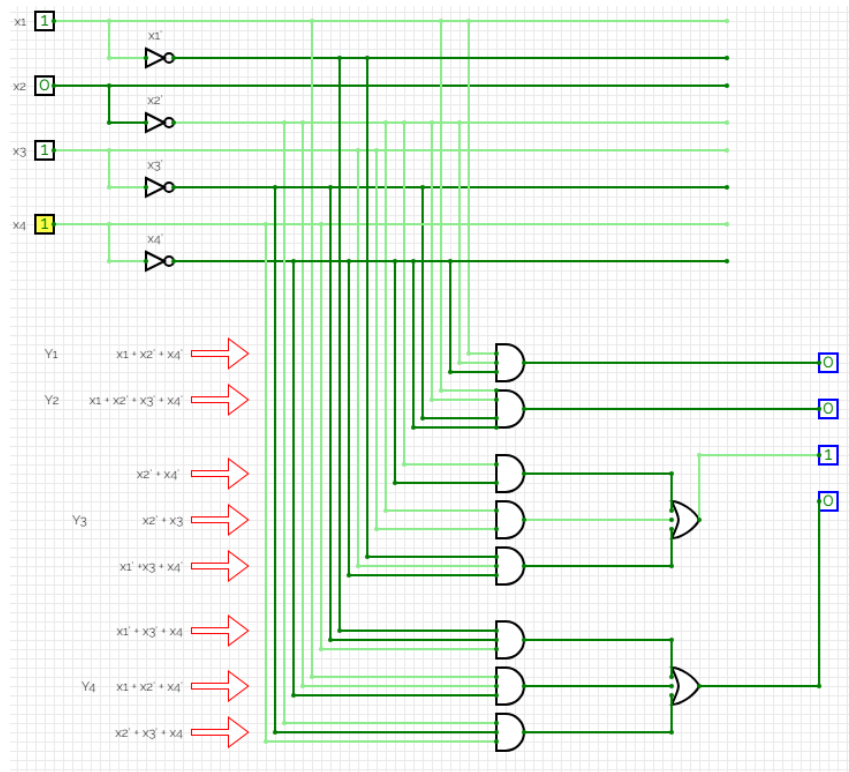


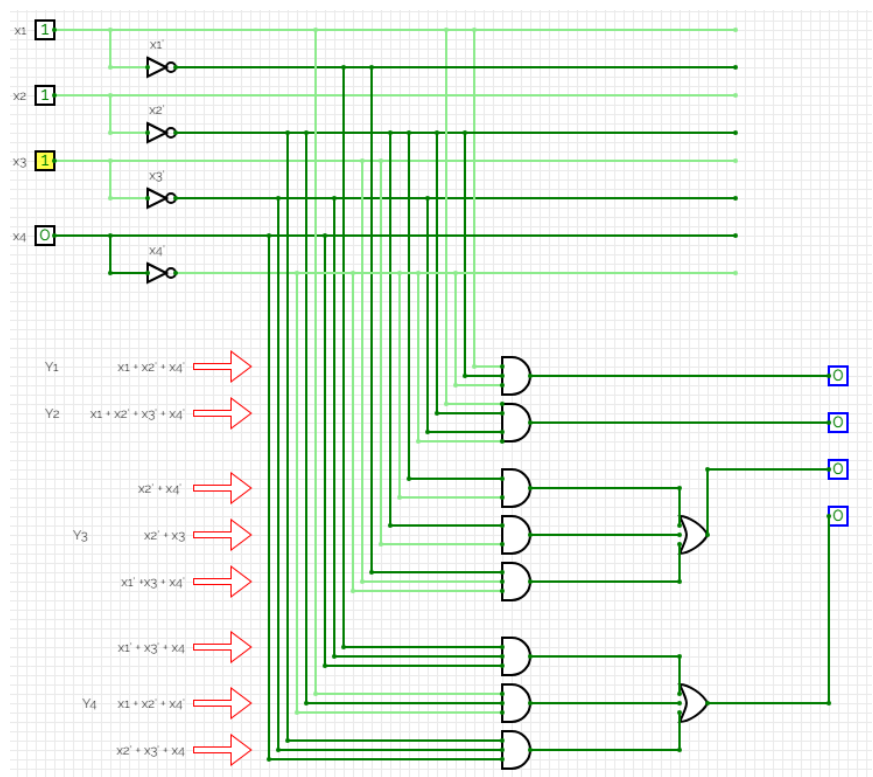
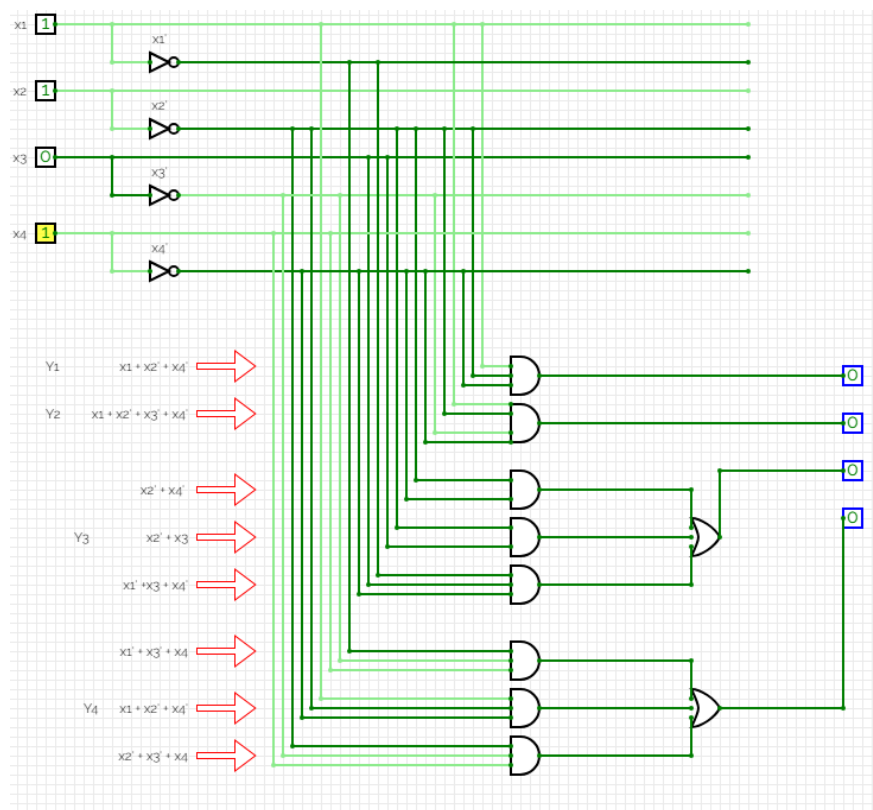


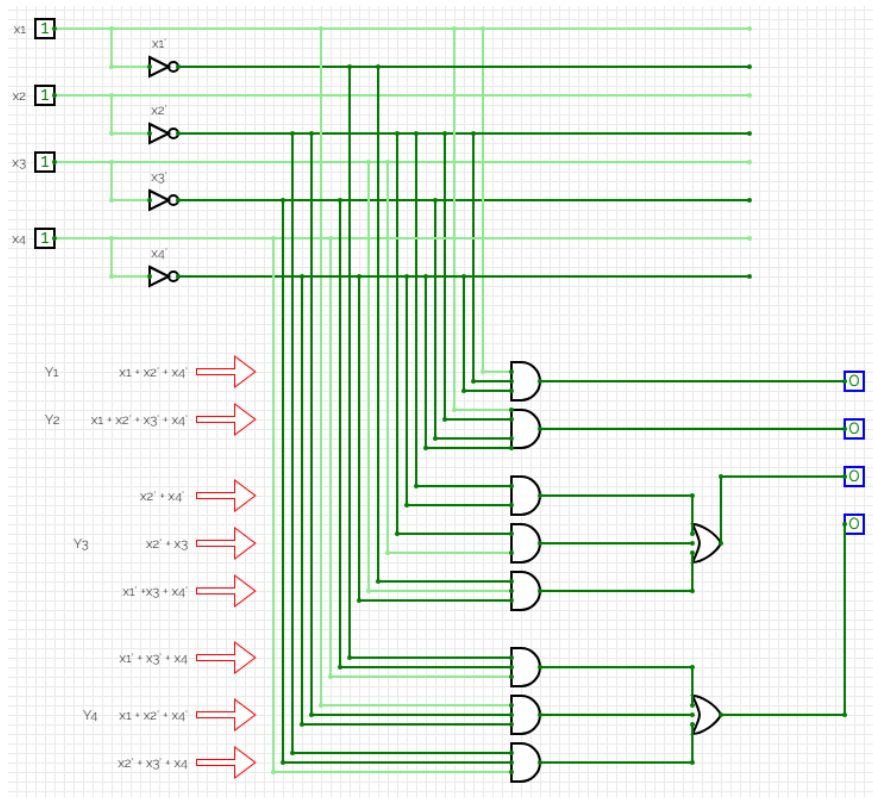












Parts List

- DIP switch for input
- 1 resistor array for DIP switch around $1k\Omega$
- 1 resistor for Bargraph LED around 500Ω
- Bargraph LED for output
- CPLD

PC/CP220 Project Phase III Checklist (3.01)

A. General

1. Professionally presented
Neat, etc. _____
2. Properly identified
(eg. name, id) _____
3. On time
at *beginning* of lab *with checklist* _____
4. Good grammar
(eg. complete sentences where required) _____
5. Correct spelling _____

B. Content

1. *Circuit diagram is
consistent with equations* _____
2. *All cases are simulated
with good explanations* _____
3. *Circuit passes/fails simulation
noted (matches truth table)* _____
4. *All input/output parts have been
listed (including resistors)* _____
5. *All previous phases included
(updated if necessary)* _____